

Lecture 9 (10/28)

As usual, we have our flow, $\varphi^t : \mathbb{R}^n \rightarrow \mathbb{R}^n, t \in \mathbb{R}^+$ which is defined as the solution of the differential equation,

$$\frac{\partial \varphi^t(x)}{\partial t} = V(\varphi^t(x)). \quad (1)$$

In the last few lectures, we have looked at the evolution of infinitesimal perturbations along the flow. Concretely, if $u(t) \in \mathbb{R}^n$ is an infinitesimal perturbation (tangent vector) at $\varphi^t(x)$, then,

$$\frac{du(t)}{dt} = \nabla V(\varphi^t(x)) u(t), \quad (2)$$

where $u(0) \in \mathbb{R}^n$ is a tangent vector at x . If $x = x^*$ is a fixed point, then, the above linear equation for the evolution of infinitesimal perturbations becomes an autonomous dynamical system:

$$\frac{du(t)}{dt} = \nabla V(\varphi^t(x^*)) u(t) = \nabla V(x^*) u(t). \quad (3)$$

For linear equations, we have already studied the stability of solutions, $u(t) = e^{tA}u(0)$, where

$$A := \nabla V(x^*).$$

This stability analysis can be summarized as follows. There is a direct sum decomposition

$$\mathbb{R}^n = E^u(x^*) \oplus E^s(x^*) \oplus E^c(x^*),$$

into the linear, invariant subspaces $E^u(x^*)$ – unstable, $E^s(x^*)$ – stable and $E^c(x^*)$ – center. An invariant subspace is one that is preserved by the dynamics. That is, if $v \in E$, then, $e^{tA}v \in E$ at all times t . By definition of a direct sum decomposition, a vector, say $v \in \mathbb{R}^n$, can be written as $v = v^u + v^s + v^c$ uniquely, with $v^u \in E^u(x^*)$, $v^s \in E^s(x^*)$ and $v^c \in E^c(x^*)$ being its unstable, stable and center components respectively. Now, these three components have different asymptotic behavior under the linear dynamics, (3). A vector $v^s \in E^s(x^*)$ goes to zero asymptotically at an exponential rate, $\alpha > 0$:

$$\|e^{tA}v^s\| \leq C e^{-\alpha t}, t \rightarrow \infty.$$

And indeed, the subspace $E^s(x^*)$ is equivalently defined by the generalized eigenspace with the corresponding eigenvalue having real part < 0 . Now, the unstable subspace, $E^u(x^*)$ is the generalized eigenspace of all the eigenvalues of A with real part > 0 .

Exercise: show that for any generalized eigenvector v with corresponding eigenvalue having real part > 0 , $\|e^{tA}v\| \geq (1/C) e^{\alpha t}$, as $t \rightarrow \infty$.

The subspace E^u is defined by asymptotic decay in norm backward in time. That is, $E^u(x^*)$ is the set of all vectors, v , that go to zero backward in time asymptotically, at an exponential rate, $\alpha > 0$,

$$\|e^{tA}v\| \leq C e^{-\alpha|t|} \|v\|, t \rightarrow -\infty. \quad (4)$$

Note that, while the vectors in E^u blow up exponentially in norm forward in time, they are not *all* the vectors that do so! A vector v blows up exponentially in norm as $t \rightarrow \infty$ as long as its unstable component, $v^u \in E^u(x^*)$ is non-zero. However, $E^u(x^*)$ really is the only set of vectors that decay exponentially in norm backward in time. Now, these two subspaces, $E^s(x^*)$ and $E^u(x^*)$ are characterized by exponential decay and growth respectively. This leaves us with the remainder, $E^c(x^*)$, in the direct sum decomposition of $\mathbb{R}^n$, which has *sub-exponential* growth under the linear dynamics. As you might guess, $E^c(x^*)$ is spanned by the generalized eigenvectors of A with real part of the eigenvalues $= 0$.

Exercise: show that for any generalized eigenvector (not regular eigenvector) v of A with corresponding eigenvalue having real part $= 0$, $\|e^{tA}v\| = o(t^k \|v\|)$, for some $k > 1$, and $t \rightarrow \infty$.

So, this is it. We now know all about asymptotic behavior of infinitesimal linear perturbations at a fixed point.

When A has no eigenvalues with zero real part, we call x^* a hyperbolic fixed point. At a hyperbolic fixed point, we have a trivial $E^c(x^*)$ – it only contains the zero vector. For a stable hyperbolic fixed point, i.e., when $E^u(x^*)$ is empty and $E^c(x^*) = 0$, we have learned a result about asymptotic stability. This is Theorem 4.19 in the book. This theorem says that linear asymptotic stability (every non-zero tangent vector is in $E^s(x^*)$) implies asymptotic stability. Asymptotic stability means that there is a neighborhood around x^* such that any x in that neighborhood goes to x^* asymptotically: $\varphi^t(x) \rightarrow x^*$ as $t \rightarrow \infty$. Broadly speaking, the asymptotic behavior of the linearization (2) predicts the *local* asymptotic behavior of the original system (1).

Even when the fixed point is not stable hyperbolic but just hyperbolic, this broad fact about the linearization predicting behavior of the original

system holds. More precisely, a homeomorphism between the flow and the linearization exists around a hyperbolic fixed point: this was the Hartman-Grobman theorem (theorem 4.36 in the book). We will now study a more general result in the same spirit: the stable and unstable subspaces have local, nonlinear extensions that are invariant under the dynamics. We will state this result – called the stable/unstable manifold theorem – and provide a proof sketch in the next lecture.

For now, we define *global* stable and unstable manifolds. These are invariant manifolds of the same dimension as their linear counterparts (E^s and E^u respectively). They are usually not embedded but only immersed in $\mathbb{R}^n$. Intuitively, this means that the function mapping points on these curves to points in $\mathbb{R}^n$ is not one-to-one (not injective), although locally this function is differentiable. For instance, a figure 8 knot in $\mathbb{R}^2$ is an immersion but not an embedding. The global stable manifold, typically denoted by $E^s(x^*)$, is the set of points whose distance to x^* decreases asymptotically to 0. That is,

$$W^s(x^*) = \left\{ x \in \mathbb{R}^n : \lim_{t \rightarrow \infty} \|\varphi^t(x) - x^*\| = 0 \right\}. \quad (5)$$

Similarly, the unstable manifold, denoted $W^u(x^*)$, is defined using backward orbits,

$$W^u(x^*) = \left\{ x \in \mathbb{R}^n : \lim_{t \rightarrow -\infty} \|\varphi^t(x) - x^*\| = 0 \right\}. \quad (6)$$

Exercise: show that the stable manifold is forward-invariant ($t \rightarrow \infty$) and the unstable manifold is backward-invariant. Give an example to show that the stable manifold is not, in general, backward invariant.

It is part of the stable/unstable manifold theorem to prove that these manifolds are locally smooth (as smooth as the flow) and tangent to E^s and E^u respectively. Note that, in class, we will only discuss this theorem at hyperbolic fixed points. But, these invariant manifolds, and the stable/unstable manifolds and the linear subspaces can all be defined along orbits, not just fixed points. We will close with their definitions in such a context.

Uniformly hyperbolic flow. The flow φ^t is called a uniformly hyperbolic flow if there exist constants $\alpha > 0, C > 0$ such that at every x , the tangent space at x , $\mathbb{R}^n \equiv T_x \mathbb{R}^n = E^u(x) \oplus E^s(x) \oplus E^c(x)$, where

- $E^u(x) := \{u(0) \in \mathbb{R}^n : u(t) \text{ solves (2), } \|u(t)\| \leq C e^{-\alpha|t|} \|u(0)\|, t \rightarrow -\infty\},$

- $E^s(x) := \{u(0) \in \mathbb{R}^n : u(t) \text{ solves (2), } \|u(t)\| \leq C e^{-\alpha t} \|u(0)\|, t \rightarrow \infty\},$
- $E^c(x) := \text{span}\{V(x)\},$

are invariant subspaces in the sense that if $u(0) \in E^s(x)$, then, $u(t) \in E^s(\varphi^t(x))$.

Exercise: show that V satisfies the linear perturbation equation, (2), and therefore, any infinitesimal perturbation along the vector field V does not exhibit growth or decay.

Now the unstable and stable manifolds are defined as before, but asymptotic convergence to an orbit defines them, not to a fixed point.

$$W^u(x) := \{y \in \mathbb{R}^n : \|\varphi^t(x) - \varphi^t(y)\| \rightarrow 0 \text{ as } t \rightarrow -\infty\}. \quad (7)$$

Similarly, the global stable manifold is defined as

$$W^s(x) := \{y \in \mathbb{R}^n : \|\varphi^t(x) - \varphi^t(y)\| \rightarrow 0 \text{ as } t \rightarrow \infty\}. \quad (8)$$

Once again, the global stable/unstable manifolds, even when it is the same at all x , are only immersed typically. They have the same dimension as the stable and unstable subspaces, but more on this later.